

Jared Marx-Kuo

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EDUCATION	PhD Student , Stanford University Math Department Expected June 2024 Bachelor of Arts in Mathematics with Honors , University of Chicago Cumulative GPA: 3.966	2019 – Present 2019
AWARDS	NSF Graduate Research Fellow , National Science Foundation • Full Funding for 3 years of PhD Stanford Graduate Fellowship in Science & Engineering , Stanford University • Full Funding for 3 years of PhD Paul R. Cohen Memorial Prize , University of Chicago • “Awarded to the graduating seniors who have achieved the highest academic record in the field of mathematics”	2021 2019 2019
TEACHING	CAAP TA , Chicago Academic Achievement Program • Teaching assistant for introduction to proofs via calculus SUMAC TA , Stanford University Mathematics Camp • Teaching Assistant for summer course in algebraic topology • Gave talk, “ <i>Math, Science, and Society</i> ” Math 171 CA , <i>Fundamental Concepts of Analysis</i> Math 62CM TA , <i>Modern Mathematics: Continuous Methods</i> LSP Math Instructor , Leland Scholars program • Instructor for course on high school level mathematics for incoming FGLI freshmen • Gave welcome talk, “ <i>Mathematics and Society</i> ”	June 2019 – August 2019 June 2020 – August 2020 September 2020 – November 2020 January 2021 – March 2021 August 2021
PUBLICATIONS AND PREPRINTS	“Variations of Renormalized Volume for Minimal Submanifolds of Hyperbolic Space” “Characters of Renner Monoids and Their Hecke Algebras” “Sandpile Groups of Cayley Graphs of $\mathbb{F}_2^N$ ” “Convolution Algebras for Finite Reductive Monoids”	
CONFERENCES AND SUMMER SCHOOLS	BIRS New Directions in Geometric Flows 2021 Virtual Conference in Geometric Analysis Seminaire de Mathematiques Superieures 2021 , MSRI Summer school in Microlocal Analysis Young Mathematicians Conference , Conference for undergraduates in math University of Chicago Summer School in Analysis , Program Website	Fall 2021 Summer 2021 Summer 2018 Summer 2016
OUTREACH	Stanford Directed Reading Program Mentor , Program Website Stanford Directed Reading Program Organizer , Program Website • Organized program and paired undergraduates with math graduate students for quarter-long, 1-on-1 reading projects • Advertise the program, schedule presentations, submit budget updates, etc. SUMAC Application Reader , Program Website • Read applications for Stanford University Mathematics Camp • Advocated for underrepresented and underresourced students in the application process SURIM Mentor , Stanford Undergraduate Research Institute in Mathematics • Mentored undergraduates in PDEs with applications to the Allen-Cahn Equation	September 2019 - Present January 2020 - Present March 2021 - April 2021 June 2020 – August 2020

SURA Mentor, [Program Website](#)

January 2020 - April 2020

- Mentored undergraduates in math by advising them on courses and summer projects

FLI Community Mentor, [Program Website](#)

January 2020 - November 2020

- Supported low income undergraduate student with navigating college life and adjusting plans for Covid-19 pandemic

CAAP RA, [Chicago Academic Achievement Program](#)

June 2019 – August 2019

- Resident Assistant for bridge program for first-gen, low income, immigrant students attending the University of Chicago

**STUDENT
SEMINARS****Student Mathematical Physics Seminar coordinator, Stanford Math Department**

- Founded Winter 2020
- *Introduction to Quantum Mechanics* (January 2020)
- *Introduction to Gauge Theory* (May 2020)

Student Analysis Seminar, Stanford Math Department

- *An Entropic View of the Central Limit Theorem* (November 2019)
- *Γ -convergence and the Allen Cahn Equation* (October 2020)
- *Ginzburg-Landau and Renormalized Energies* (November 2020)
- *Energy Methods in Linear Wave Equations - An Intro to Hyperbolic PDES* (February 2021)
- *Introduction to Optimal Transport and Coupling* (April 2021)
- *Introduction to Renormalized Volume* (November 2021)

Student Probability Seminar, Stanford Math Department

- *Hypercontractivity and Log-Sobolev Inequalities* (October 2019)

Stanford “Kiddie Colloquium”, Stanford Math Department

- *Hypercontractivity and Log-Sobolev Inequalities* (November 2021)

REFERENCES**Rafe Mazzeo**, Professor, Department of Mathematics, Stanford, rmazzeo@stanford.edu**Otis Chodosh**, Professor, Department of Mathematics, Stanford, ochodosh@stanford.edu